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# Robust Combination Testing: Methods and Application to COVID-19 Detection

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**Abstract.** Simple and affordable testing tools are often not accurate enough to be operationally relevant. For coronavirus disease 2019 (COVID-19) detection, rapid point-of-care tests are cheap and provide results in minutes, but they largely fail policy makers' accuracy requirements. We propose an analytical methodology, based on robust optimization, that identifies optimal combinations of results from cheap tests for increased predictive accuracy. This methodological tool allows policy makers to credibly quantify the benefits from combination testing and thus, break the trade-off between cost and accuracy. Our methodology is robust to noisy and partially missing input data and incorporates operational constraints—relevant considerations in practice. We apply our methodology to two data sets containing individual-level results of multiple COVID-19 rapid antibody and antigen tests, respectively, to generate Pareto-dominating receiver operating characteristic curves. We find that combining only three rapid tests increases out-of-sample area under the curve by 4% (6%) compared with the best-performing individual test for antibody (antigen) detection. We also find that a policy maker who requires a specificity of at least 0.95 can improve sensitivity by 8% and 2% for antibody and antigen testing, respectively, relative to available combination testing heuristics. Our numerical analysis demonstrates that robust optimization is a powerful tool to avoid overfitting, accommodate missing data, and improve out-of-sample performance. Based on our analytical and empirical results, policy makers should consider approving and deploying a curated combination of cheap point-of-care tests in settings where "gold standard" tests are too expensive or too slow.

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**Keywords:** diagnostic operations • combination testing • knapsack • robust optimization • healthcare analytics

## 1. Introduction

In diagnostic testing, there is a fundamental trade-off between diagnostic accuracy on the one hand and testing cost and speed on the other hand. For coronavirus disease 2019 (COVID-19), for instance, the laboratory-based "gold standard" reverse transcription polymerase chain reaction (RT-PCR) test for detection of current infection requires expensive laboratory capacity and a supply chain connecting testing locations and laboratories, which increase both processing time and costs. In contrast, point-of-care (POC) lateral flow antigen tests are cheap and provide results in minutes, but they largely fail policy makers' accuracy criteria (Wise 2020). The same dilemma is faced in the detection of acquired immunity after infection, where lateral flow antibody tests offer lower accuracy than laboratory tests. This drawback is exacerbated in the early phases of the spread of a novel pathogen, when the first tests available may be of lower quality. The inferior quality of POC tests has greatly limited their use for COVID-19 (Shuren and Stenzel 2021) and has resulted in millions wasted on purchasing tests subsequently deemed unfit for use (Armstrong 2020).

Despite these shortcomings, experts argue that frequent, fast, and large-scale antigen testing schemes, for which RT-PCR tests are clearly unsuitable, should be implemented to enable the safe reopening of economic activity (Mina et al. 2020, Larremore et al. 2021), especially in use cases involving entry into public spaces, such as airports, care homes, universities, and entertainment or sporting venues. Similarly, efficient and widespread antibody testing is useful to assess global immunity and prioritize vaccination delivery (Bubar et al. 2021). Beyond the specific case of COVID-19 diagnostics, the usefulness of combining multiple cheap diagnostic options to improve overall accuracy extends

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to other healthcare contexts as well as to applications such as maintenance or recruitment, where results of multiple diagnostic or classification methods are aggregated to make a single recommendation.

In this paper, we propose an optimization-based methodology to combine the results of different POC tests to achieve higher accuracy without compromising affordability or convenience. Intuitively, there are potentially both systematic and stochastic mechanisms that motivate the idea of combining results obtained from multiple tests. First, different tests can use different technologies or detection strategies (e.g., the target protein, chemical reagents used, or how the tests are administered may differ). In this case, each test provides a different perspective on a patient's underlying condition, and combining them will lead to a more comprehensive and accurate picture. Second, test results are inherently random because of uncertainty in test administration, chemical processes, and unobserved patient characteristics. In this case, if the correlation between test outcomes is low, combining them can help reduce noise and improve accuracy. For both of these reasons, we expect that combining tests can improve accuracy. For a stylized example of the systematic or stochastic mechanisms driving the benefits of combining tests, see the appendix.

Any methodology for combining multiple test outcomes must overcome three fundamental challenges in the design of combination testing policies. The first is how to interpret an *outcome sequence* of discordant yet potentially correlated test results (e.g., $(+, -, +)$ in the simple case of combining three tests) to classify a person as positive or negative. Heuristics, such as classifying an outcome sequence as positive only if all the test outcomes are positive, are simple but can have suboptimal performance.[1] The second challenge is to select which tests to combine when multiple tests are available. A naïve approach of combining the tests with highest individual accuracy may result in suboptimal outcomes if their results are not complementary when combined. Test selection must also account for operational and budget constraints, affecting how many tests can feasibly be combined in a given setting. The third challenge is ensuring an empirical evidence base for the proposed combination testing policy—by using available data on the performance of individual tests—while avoiding overfitting, which would result in poor out-of-sample classification and therefore, limit applicability.

Analytically, we formulate the policy maker's problem as one of selecting and combining a subset of available POC tests to maximize diagnostic accuracy measured by sensitivity and specificity (i.e., the probability that a condition-positive or condition-negative individual is classified as positive or negative, respectively). First, for a given set of tests that are used in combination, we prove that a simple ranking procedure identifies all classification rules with Pareto-optimal accuracy (Section 3.1) by formulating this problem as a version of the well-studied knapsack problem. The resulting set of optimal classification rules yields a dominating receiver operating characteristic (ROC) curve—with the highest attainable area under the curve (AUC). Of potentially independent interest, our proof provides an optimization-based derivation of the Neyman–Pearson lemma (McIntosh and Pepe 2002). Note that our approach is nonparametric and does not require any assumptions about the correlation structure among tests results. Second, we extend our formulation to incorporate the selection of tests to be included in the combination while accounting for the cost of adding each test and the policy maker's overall budget (Section 3.2). Third, we present a robust variant of the model to account for noise in the input parameter estimates, such as the sensitivity and specificity of individual POC tests (Section 3.3). Indeed, in practice, the data available for calibrating these parameters are often limited, leading to finite-sample estimation error. For emergency use authorization, for instance, manufacturers can use as few as 30 positive samples (Fitzpatrick et al. 2021), which can result in overoptimistic estimates of test performance (Armstrong 2020). Our robust model prevents such overfitting, and we provide theoretical guarantees on its out-of-sample diagnostic performance. Fourth, data gathered for manufacturer-independent head-to-head comparisons of tests often contain a significant number of missing entries because all tests are not necessarily applied on all samples. Importantly, we extend our robust formulation so that it can be directly applied on data with missing entries (Section 3.4).

Empirically, we apply the four steps of our methodology to two data sets shared with us by medical research institutions in the United Kingdom and Germany, which record the performance of multiple commercially available point-of-care antibody (Adams et al. 2020) and antigen (Corman et al. 2021) tests for COVID-19 (Sections 5.1–5.4). Through this analysis, we make several contributions. First, we demonstrate that optimally combining multiple tests using our robust optimization method improves diagnostic accuracy compared with using single tests or standard heuristics for combining tests. Notably, we find that combining three tests achieves a 4% (6%) relative improvement in out-of-sample AUC for antibody (antigen) testing compared with using the best individual test. We also find that if a policy maker requires specificity of at least 0.95, our methodology achieves 8% and 2% relative improvements in sensitivity for antibody and antigen detection, respectively, compared with standard heuristics. Second, our analysis demonstrates that with limited available data, the risk of overfitting is significant, an issue that is efficiently mitigated by our robust formulation. Similarly, regarding missing data, incorporating an imputation step into

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our robust combination algorithm substantially outperforms policies generated using maximum-likelihood (ML) imputation in out-of-sample comparisons. Finally, we provide an illustrative example of how a decision maker who is responsible for designing testing protocols in a given community (e.g., for managing access to public spaces or outbreak detection) would apply our methodology to develop a testing policy, even in the realistic situation where data are limited.

Our work has direct and immediate policy implications. We introduce an actionable method for selecting and combining POC tests to achieve superior diagnostic accuracy. For detecting COVID-19 infection, a combination of two or three POC tests is likely to be cheaper and faster than RT-PCR, which requires both a laboratory infrastructure and a sample transportation supply chain (Rubin 2020, Böger et al. 2021). Our method offers policy makers a tool to credibly quantify the benefits from combination testing and quickly identify accurate combinations of tests if they exist. In addition, our combination testing approach delivers an entire ROC curve, which allows policy makers to emphasize either sensitivity or specificity using the same combination of tests depending on their relative priorities for a particular population. Given our results, policy makers should consider approving and deploying combinations of tests in settings where RT-PCR testing is too costly or too slow but where individual POC tests do not provide enough diagnostic accuracy. Although directly implementable by performing multiple tests made by different manufacturers, combination testing can also be implemented by creating a single diagnostic device that combines results from multiple tests "under the hood."

From a research standpoint, we believe that this is the first paper to explore the value of optimally combining multiple diagnostic tests through analytical modeling and empirical analysis. The medical literature has considered simple heuristics for combining a given set of tests (Marshall 1989, Sox et al. 2013), which can be suboptimal when combining even as few as three tests. Our method not only identifies *how to* combine tests but also *which* subset of tests to combine without conducting an exhaustive search. Furthermore, we provide much needed empirical evidence on the potential benefits of combination testing. Although we illustrate our findings on data from COVID-19 testing, our model development is general and provides an application-agnostic methodology for combining the results of multiple tests. Therefore, our methodology can be applied in any setting where data on different classifiers are collected on the same subject, be it for recruitment purposes or machine maintenance. Our analysis demonstrates the value of structuring data collection such that it allows for both head-to-head comparison of different tests as well as the development of combination testing policies.

## 2. Literature Review

Our key contribution is to the ongoing debate on how to expand COVID-19 testing in a cost-effective way (Section 2.1). Our analysis also contributes to the operations management literature on diagnostics, clinical decision making, and operational innovations to battle COVID-19 (Section 2.2). More generally, our analysis relates to the machine learning literature on combining predictions for improved classification accuracy (Section 2.3) as well as prior work on the predictive power of using multiple tests in the context of test sequencing (Section 2.4).

### 2.1. COVID-19 Testing

The "gold standard" test for COVID-19, RT-PCR (Böger et al. 2021), requires laboratory equipment, which increases the cost per test and induces a sample-to-results delay of the order of 24–48 hours (Larremore et al. 2021). In addition, molecular laboratories in some countries have sometimes stopped accepting samples for RT-PCR tests during demand surges because of manpower shortages (*The Economic Times* 2021).

At the same time, a plethora of POC tests have been developed for COVID-19 testing for both antibody and antigen identification. In fact, the U.S. Food and Drug Administration has approved over 200 different POC diagnostic tests for COVID-19 since the start of the pandemic (Food and Drug Administration 2020). The majority of POC tests for COVID-19 rely on lateral flow technology. Cost estimates for this type of POC test (often around six dollars) vary across studies and range from one third to one tenth the cost of an RT-PCR test (Rubin 2020, Böger et al. 2021). In addition, some progress has also been made in bringing the RT-PCR technology closer to the point of care. For example, portable devices that achieve high sensitivity have been deployed at hospitals to alleviate pressure on laboratories (Gibani et al. 2020). However, these devices do not eliminate the need for expensive reagents and often need to be operated by professionals (Gupta et al. 2021). The methods presented in this paper would apply to a broad range of diagnostic technologies.

Governments have committed significant funds to expand diagnostic testing during COVID-19, including POC testing. The United States has pledged $50 billion for this purpose (Politico 2021), and the United Kingdom has allocated £22 billion to its national testing program (National Audit Office 2020). However, the use of POC tests in the battle against COVID-19 is widely debated. Deeks et al. (2021) argue that POC testing can be useful but only if highly accurate. They summarize trial results from the United Kingdom and conclude that the potential harm of false negatives outweighs the potential benefits. In contrast, Mina et al. (2020) argue in favor of POC test deployment, particularly for routine

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testing of asymptomatic individuals, to maximize the overall sensitivity of the testing regime, despite potentially low sensitivity for each individual.

Our methodological contribution and its empirical validation present a way to improve the accuracy of POC tests in a cost-efficient manner. Our data-driven approach reveals that the combination of even two POC tests can significantly improve diagnostic accuracy over that of an individual POC test while remaining much faster and cheaper than RT-PCR testing.

It is worth noting that the overall cost of a test depends on a number of factors, including volume ordered, public subsidies, method of administration, and market competition. For RT-PCR tests, for example, there is a wide variability in prices within and between countries (International Air Transport Association 2021). There is also a lack of reliable and objective data on test costs. In a comparison of test prices among the 566 top U.S. hospitals, for example, Xiao and Rathi (2021) note that two thirds disclosed none.

### 2.2. Diagnostic Operations and Clinical Decision Making

The clinical decision-making literature has explored the utility and cost-effectiveness of conducting additional tests (Weinstein and Fineberg 1980, Hershey et al. 1986, Sox et al. 2013) by considering simple rules of thumb (e.g., classifying an outcome as positive if any test is positive (the `any` rule), if all tests are positive (the `all` rule), or if the majority of tests are positive (the `majority` rule)). This literature examines the net utility from choosing each of the available courses of action by taking as inputs a set of available tests; the diagnostic accuracy of each test; a set of alternative decisions (e.g., "conduct two tests in parallel and treat if both are positive"); and the utility associated with each course of action. However, no prior work provides a systematic methodology for classifying the outcome sequences of $m$ tests for best achievable sensitivity and specificity. A key contribution of our work is to develop such a method and demonstrate its effectiveness on real data. Using our method, we identify the set of classification rules that dominate all other rules in terms of sensitivity and specificity for a given combination of tests. This set of best achievable classification rules for a given set of tests can be summarized in a dominating ROC curve.

In the operations management literature, the value of increasing testing accuracy is ambiguous. Although performing additional tests can improve diagnostic accuracy, hence benefiting downstream decision making, it can also generate additional delays. Early modeling work in the context of maintenance combines maintenance interventions with inspections to account for stochastically failing equipment (McCall 1965, Wang 2002). Alizamir et al. (2013) analyze the trade-off between improving testing accuracy (e.g., by performing additional examinations) and treatment delays. They present an optimal policy that depends on prior beliefs about the patient type and the congestion level of the system. Kremer and de Véricourt (2022) conduct controlled laboratory experiments to assess decision makers' biases when balancing accuracy and congestion. Linking operational implementation and testing, Levi et al. (2019) study the scheduling trade-off between task diagnosis and task processing, and Aprahamian et al. (2019) develop an optimal scheme for testing groups of patients using an imperfect single test, assuming subject-specific risk characteristics.

Recent papers have examined the operational implementation of testing to curb the spread of COVID-19. Several have examined allocation of limited testing capacity (e.g., to health centers in the Philippines (Buhat et al. 2021), to patients with different probability of being infected (Calabrese and Demers 2022), or to balance the immediate benefit of diagnosing infected individuals with the benefit of learning population-level infection rates (Kasy and Teytelboym 2020)). Bastani et al. (2021) propose and implement a bandit framework to specifically target the allocation of tests to incoming international travelers at the Greek border. Yang et al. (2022) develop a game-theoretic queueing model to identify improved scheduling and pricing policies for COVID-19 testing facilities. Pulia et al. (2020) propose a multitiered sequential testing policy in which patients are first triaged at the point of care using rapid antigen tests before using RT-PCR technology for diagnostic confirmation. Drakopoulos and Randhawa (2021) consider tests as information products with limited availability and demonstrate that if targeted allocation of tests is impossible, moderately accurate tests are preferable in terms of social outcome. Dai and Singh (2023) explore the risk of overtesting in situations where laboratories can decide on the diagnostic criterion (which controls sensitivity) to apply for RT-PCR testing. These papers focus on the operational implementation of a given testing policy with a given diagnostic accuracy. In contrast, we propose a methodology for improving diagnostic accuracy.

In a conceptual article, Ramdas et al. (2020) suggest using multiple tests with independent outcomes to increase the accuracy of COVID-19 testing. In this paper, we develop the methodology required to analyze data on the comparative performance of tests with correlated outcomes and identify powerful and cost-effective combination strategies using data from two medical institutions.

### 2.3. Combining Predictions for Improved Accuracy

Our analysis relates to aggregation methods in machine learning—particularly *stacking* and *bagging*. The idea of combining predictions from different models, originally

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proposed by Wolpert (1992) and referred to as *stacking* (see Džeroski and Ženko 2004 for a review), is widely used for binary classification. For instance, Barreno et al. (2007) leverage the Neyman–Pearson lemma (Neyman and Pearson 1933) to optimally combine binary predictions obtained from different classifiers by ranking outcome sequences by their likelihood ratios. In medical research (and cancer screening in particular), the lemma has been applied to improve diagnostic accuracy by combining different biomarkers (McIntosh and Pepe 2002). Our method recovers the likelihood ratio ranking rule using an optimization rather than statistical lens. In practice, however, each test is costly, and combining all of them would be impractical. Our optimization perspective naturally accounts for such operational constraints and enables identification of the best subset of tests to be combined together, without exhaustive enumeration.

Breiman (1996) proposed bootstrap aggregation or *bagging*, in which weak classification algorithms (referred to as base learners) are trained on bootstrap samples of training data and then aggregated into a single prediction (usually via a heuristic decision rule, such as classifying according to the majority of base learners or averaging across them). Our work shares the same aggregation flavor. However, unlike bagging, our base learners are tests developed by different manufacturers and not obtained on bootstrap samples. Hence, we learn the optimal aggregation policy directly from data on test performance without any knowledge about the data-generating process for each individual index test.

Finally, overfitting and out-of-sample performance are important practical issues in our setting, which the standard Neyman–Pearson lemma and the likelihood ratio ranking rule (McIntosh and Pepe 2002, Barreno et al. 2007) do not account for. In statistics, robust versions of the lemma have been developed to account for distribution misspecification (Huber 1965, Huber and Strassen 1973) but have not been used in the context of combining tests. We formulate a robust knapsack optimization problem to directly mitigate the impact of finite sample estimation error and missing data on combination testing while accounting for operational constraints. Compared with previous work on robust knapsack problems with an uncertain objective (Monaci and Pferschy 2013, Cheng et al. 2014), uncertainty affects both the objective and the constraints of our problem.

### 2.4. Sequencing of Tests

In medical decision making, there is a broad literature on contingent testing strategies. In this approach, patients are screened with a first test. If the result is positive, then the final diagnosis is confirmed with a second test. These sequencing strategies have been studied, for instance, in the context of mammography examinations (Alagoz et al. 2013) and COVID-19 (Paltiel et al. 2020, Smith et al. 2021). For this to be useful in practice, the second (confirmatory) test should be more reliable than the first one, which usually means that it is also more expensive or invasive. In the context of COVID-19 detection, this would correspond to first performing a rapid antigen test and confirming any positive result with RT-PCR (e.g., as piloted in the United Kingdom in Liverpool) (Wise 2020).

One insight from this literature is that the first-step "screening" test needs to be accurate enough for these policies to be effective. A low false-negative rate ensures that condition-positive individuals are detected. A low false-positive rate ensures that the second (expensive) test is not wasted on many condition-negative individuals. However, with COVID-19 antigen tests, many have argued that they do not meet such standards (Wise 2020). With regard to this literature, our key contribution is to propose a solution to improve the accuracy of the first-step test with little compromise on their price so that such contingent testing strategies can be effective in practice. We also believe that there are use cases, such as entry to public places, where one cannot afford a second-step confirmation test because of monetary or time considerations. Therefore, improving the accuracy of cheap and fast tests is of independent interest.

## 3. Model

In this section, we present a methodology for combining multiple tests for improved diagnostic accuracy. We first demonstrate that the problem of optimally classifying outcome sequences from multiple tests can be formulated as a knapsack problem (Section 3.1). Under this lens, we provide a new proof of the Neyman–Pearson lemma from first principles. We then extend the knapsack model for a given set of tests to the problem of selecting which tests to include in the combination, subject to a budget constraint (Section 3.2). Finally, we present a robust version of the model to account for in-sample estimation error (Section 3.3) and missing data (Section 3.4). Additionally, Section EC2 in the electronic companion provides a detailed optimization formulation for how to sequence tests once the policy has been determined.

In the following, we use lowercase nonboldfaced ($x$), lowercase boldfaced ($\boldsymbol{x}$), uppercase boldfaced ($\boldsymbol{X}$), and uppercase script-faced ($\mathcal{X}$) characters to denote scalars, vectors, matrices, and sets, respectively. We use hats to denote sample estimates of population parameters. For $\boldsymbol{x} \in \mathbb{R}^n$, its $i$th component is denoted $x_i$, and $\boldsymbol{e}$ is the vector of all ones.

### 3.1. A Knapsack Perspective on Classifying Outcome Sequences

Let $\boldsymbol{s}$ denote an outcome sequence arising from performing $m$ tests. For instance, when combining two tests, $\boldsymbol{s}$

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can be one of $2^2 = 4$ possibilities: (0, 0), (1, 0), (0, 1), (1, 1). Let $\mathcal{S}_m$ denote the set of all possible outcome sequences given $m$ tests, and let $n = |\mathcal{S}_m| = 2^m$. We identify $\mathcal{S}_m$ as $\{1, \ldots, n\}$.

Let $\hat{a}_s$ ($\hat{b}_s$) denote the fraction of condition-positive (condition-negative) individuals in the data sample with outcome sequence $\boldsymbol{s}$. Observe that $\hat{\boldsymbol{a}}$ ($\hat{\boldsymbol{b}}$) has $n = 2^m$ coordinates and fully characterizes the joint distribution of the $m$ test results on the condition-positive (condition-negative) individuals in a nonparametric manner. A classification rule is a vector $\boldsymbol{x} \in [0,1]^n$, where $x_s$ indicates the fraction of samples with outcome sequence $\boldsymbol{s}$ classified as *positive*. If $x_s = 1$ (0), the rule classifies outcome sequence $\boldsymbol{s}$ as *positive* (*negative*). For instance, in the case of two tests, the and rule, in which only individuals with both tests positive are classified as positive, corresponds to

$$x_{(1,1)} = 1 \text{ and } x_s = 0 \text{ for } \boldsymbol{s} \in \{(0,0), (1,0), (0,1)\}.$$

We use the term *classification* rule for $\boldsymbol{x}$ on purpose because the problem can be viewed as a binary classification problem of predicting the underlying condition given the outcome sequence $\boldsymbol{s}$. For a given rule $\boldsymbol{x}$, its sensitivity is given by $\sum_s \hat{a}_s x_s$, and its specificity is given by $1 - \sum_s \hat{b}_s x_s$. Given a target specificity level $t_0 \in [0,1]$, the rule $\boldsymbol{x}$ that maximizes sensitivity while achieving a specificity of at least $t_0$ is the solution of the knapsack problem:

$$\max_{\boldsymbol{x} \in [0,1]^n} \hat{\boldsymbol{a}}^\top \boldsymbol{x} \quad \text{s.t.} \quad \hat{\boldsymbol{b}}^\top \boldsymbol{x} \le (1 - t_0). \tag{1}$$

From knapsack theory (Dantzig 1957), we immediately derive the optimal solution of (1).

**Lemma 1.** *The solution obtained by sorting outcome sequences by decreasing values of $\hat{a}_s/\hat{b}_s$ and greedily filling the knapsack is optimal for* (1).

In the context of disease screening, the Neyman–Pearson lemma implies that ordering outcome sequences according to their likelihood ratios is optimal (McIntosh and Pepe 2002). Indeed, consider a person whose outcome sequence is $\boldsymbol{s}$ and the null hypothesis that this person is condition negative. In this case, setting $x_s = 1$ is analogous to rejecting the null hypothesis. The Neyman–Pearson lemma states that the likelihood ratio rule, which rejects the null hypothesis whenever the likelihood ratio $\hat{a}_s/\hat{b}_s$ exceeds some desired threshold, is the uniformly most powerful test. In particular, this ordering and the resulting optimal classification rule do not depend on the prevalence of the disease. Lemma 1 recovers this simple result (see McIntosh and Pepe 2002, section 3) yet by invoking tools from optimization rather than statistics. This new perspective lends itself to natural extensions to account for operational and statistical estimation issues that we will present in the rest of this section.

In Problem (1), we did not constrain the components of $\boldsymbol{x}$ to be binary. If $x_s \in (0,1)$, the rule is no longer deterministic, and a person with outcome $\boldsymbol{s}$ will be classified as *positive* with probability $x_s$. However, Lemma 1 shows that there always exists an optimal solution with at most one fractional entry. Also, for certain specific values of $t_0$, the optimal rule is integral. In the remainder of the paper, we focus our attention on binary vectors $\boldsymbol{x} \in \{0,1\}^n$ (i.e., deterministic mappings from sequences $\boldsymbol{s} \in \mathcal{S}_m$ to a positive/negative classification). Indeed, probabilistic classification rules, although theoretically feasible, could prove harder to implement in practice.

By solving (1) for different values of $t_0$ (as described in Algorithm 1 in Section EC4 in the electronic companion), we construct a piecewise linear ROC curve that represents the set of Pareto-dominating classification rules.

### 3.2. Accounting for Operational Constraints

We leverage the knapsack optimization perspective from Section 3.1 to extend the analysis to account for cost and operational constraints. To do so, we introduce a binary variable $y_k \in \{0,1\}$ indicating whether test $k = 1, \ldots, m$ is used by the classification rule $\boldsymbol{x}$. Budget constraints can then be expressed linearly as $\sum_{k=1}^m c_k y_k \le K$, where $c_k$ is the cost of each individual test $k$ and $K$ is the per-person total allowable budget.

We capture the logical relationship between $\boldsymbol{x}$ and $\boldsymbol{y}$ via linear constraints. For every outcome sequence $\boldsymbol{s}$, denote by $\bar{\boldsymbol{s}}^k$ the outcome sequence obtained by switching the result of test $k$ (i.e., $\bar{s}_j^k = s_j$ for $j \ne k$ and $\bar{s}_k^k = 1 - s_k$). For a given classification rule $\boldsymbol{x}$, if $|x_s - x_{\bar{s}^k}| = 0$, then the results of the $m-1$ other tests being equal, the result of test $k$ does not impact the classification. If this is the case for all sequences $\boldsymbol{s}$, then test $k$ is not used, and we should have $y_k = 0$. In other words, we have the following constraints:

$$\frac{1}{n} \sum_{s \in \mathcal{S}_m} |x_s - x_{\bar{s}^k}| \le y_k \le \sum_{s \in \mathcal{S}_m} |x_s - x_{\bar{s}^k}|.$$

The first inequality enforces the constraint "$y_k = 0 \Rightarrow \forall \boldsymbol{s} \in \mathcal{S}_m, x_s = x_{\bar{s}^k}$," whereas the second enforces the reverse implication. As a result, we consider combination rules in the following set:

$$\mathcal{X} := \left\{ \boldsymbol{x} \in \{0,1\}^n \;\middle|\; \exists \boldsymbol{y} \in \{0,1\}^m : \begin{array}{l} \boldsymbol{c}^\top \boldsymbol{y} \le K, \\ \frac{1}{n} \sum_{s \in \mathcal{S}_m} |x_s - x_{\bar{s}^k}| \le y_k, \ \forall k \\ y_k \le \sum_{s \in \mathcal{S}_m} |x_s - x_{\bar{s}^k}|, \quad \forall k \end{array} \right\}.$$

Additional constraints can be added to the definition of $\mathcal{X}$ to account for multiple resources or enforce some structure on the classification rule $\boldsymbol{x}$. For instance, in Section EC1 in the electronic companion, we propose adding linear constraints on $\boldsymbol{x}$ and $\boldsymbol{y}$ to force the aggregation rule $\boldsymbol{x}$ to correspond to one of the available heuristic rules (the `any`, `all`, or `majority` rule). Tests

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might also follow different protocols (e.g., saliva tests versus nasal swab), and we can add constraints on $\boldsymbol{y}$ to capture patient preferences (e.g., a limit on the number of nasal swabs required).

The modeling of testing costs allows for heterogeneous costs across tests to be included in the analysis. For simplicity and because of the absence of reliable data on test costs (Xiao and Rathi 2021), we will set $c_k = 1$ in the implementation of the approach for our empirical results. Setting $c_k = 1$ for all $k$ restricts the number of tests being used to at most $K$. As a result, our empirical analysis focuses on simultaneously selecting $K$ of $m$ tests to combine ($\boldsymbol{y}$) and the decision rule for that combination ($\boldsymbol{x}$). In terms of terminology, we refer to $\boldsymbol{x}$ as the "(classification) rule" and to $\boldsymbol{y}$ as the "test combination," whereas we use the term "policy" to refer to both the tests selected and how their results are to be interpreted.

### 3.3. Accounting for Noisy Estimation

Problem (1) uses in-sample estimates $\hat{a}_s$ and $\hat{b}_s$ and is not protected against estimation error. Our optimization perspective enables us to use robust optimization to address this concern.

First, we construct an *uncertainty set* around $\hat{\boldsymbol{a}}$, denoted as $\mathcal{U}_{\hat{a}}$, which describes a set of plausible values for $\boldsymbol{a}$. Then, we replace the sensitivity term $\hat{\boldsymbol{a}}^\top \boldsymbol{x}$ in the objective of (1) by its the worst-case value over all $\boldsymbol{a} \in \mathcal{U}_{\hat{a}}$: that is, solve

$$\max_{x \in \mathcal{X}} \left[ \min_{a \in \mathcal{U}_{\hat{a}}} \boldsymbol{a}^\top \boldsymbol{x} \right] \text{ s.t. } \hat{\boldsymbol{b}}^\top \boldsymbol{x} \le 1 - t_0. \tag{2}$$

For clarity of exposition, we detail our approach to sensitivity estimation (based on $\hat{\boldsymbol{a}}$) only but also implement a conceptually similar approach for specificity (based on $\hat{\boldsymbol{b}}$).

We decompose the constraints defining $\mathcal{U}_{\hat{a}}$ into three categories: (i) $\boldsymbol{a}$ should define valid probabilities on $\mathcal{S}_m$, (ii) the components of $\boldsymbol{a}$ should be "close" to their in-sample estimates $\hat{\boldsymbol{a}}$, and (iii) the test sensitivities computed from the components of $\boldsymbol{a}$ should be consistent with confidence intervals (CIs) on the performance of each test. We formalize these constraints.

First, $\boldsymbol{a}$ is a vector of probabilities: $\boldsymbol{a} \ge 0$ and $\sum_{s \in \mathcal{S}_m} a_s = \boldsymbol{e}^\top \boldsymbol{a} = 1$.

Second, because $\hat{\boldsymbol{a}}$ are in-sample estimates of the underlying probabilities, any candidate probability vector $\boldsymbol{a}$ could be different yet close to $\hat{\boldsymbol{a}}$. Accordingly, we bound the total absolute deviation $\sum_s |a_s - \hat{a}_s|$ with the constraint $\sum_s |a_s - \hat{a}_s| \le \Gamma$, where $\Gamma$ controls the level of robustness (i.e., protection against noise). Observe that total absolute deviation constraints are a special case of $\phi$-divergence constraints (Ben-Tal et al. 2013).[2] In our experiments, we treat the robustness parameter $\Gamma$ as a hyperparameter and calibrate its value using crossvalidation (see Section 4.3).

Third, for each test $k \in \{1, \ldots, m\}$ and any $\boldsymbol{a}$, $\sum_{s:s_k=1} a_s$ is the associated sensitivity of test $k$. Assume that $[\hat{p}_k - \hat{\rho}_k, \hat{p}_k + \hat{\rho}_k]$ defines a $(1-\alpha)$ confidence interval on the estimated sensitivity of test $k$.

**Assumption 1.** *The true probability vector $\boldsymbol{a}_0$ satisfies each constraint $|\sum_{s:s_k=1} a_{0,s} - \hat{p}_k| \le \hat{\rho}_k$ for $k = 1, \ldots, m$ with probability at least $1 - \alpha$.*

Accordingly, it is reasonable to expect $\boldsymbol{a}$ to satisfy similar constraints:

$$\left| \sum_{s:s_k=1} a_s - \hat{p}_k \right| \le \hat{\rho}_k.$$

In practice, the values of $\hat{p}_k$ and $\hat{\rho}_k$ could be provided by the manufacturer or estimated from data.

Observe that the second and third categories of constraints described serve different purposes. The second group of constraints controls the deviation of $\boldsymbol{a}$ from $\hat{\boldsymbol{a}}$ in absolute terms, whereas in the third group of constraints, an underestimate in one component of $\boldsymbol{a}$ could compensate for an overestimate in another. Furthermore, the bound $\sum_s |a_s - \hat{a}_s| \le \Gamma$ weights all components of $\boldsymbol{a}$ equally, whereas the sensitivity constraints exploit a particular problem structure (i.e., the set $\{s : s_k = 1\}$, which corresponds to test $k$ being positive). By leveraging the fact that we can reconstruct the individual test sensitivities (via partial summations), we notably reduce the size of the uncertainty set (hence, the conservatism of the approach).

Taken together, the uncertainty set $\mathcal{U}_{\hat{a}}$ is defined as

$$\mathcal{U}_{\hat{a}} := \left\{ \boldsymbol{a} \ge 0 : \boldsymbol{e}^\top \boldsymbol{a} = 1;\ \sum_{s \in \mathcal{S}_m} |a_s - \hat{a}_s| \le \Gamma;\ \left| \sum_{s:s_k=1} a_s - \hat{p}_k \right| \le \hat{\rho}_k,\ \forall k \right\},$$

and Problem (2) returns the combination rule with the best worst-case sensitivity. Under Assumption 1, our pessimistic estimate $\min_{a \in \mathcal{U}_{\hat{a}}} \boldsymbol{a}^\top \boldsymbol{x}$ provides a valid lower bound on the true performance of a combination rule, $\boldsymbol{a}_0^\top \boldsymbol{x}$.

**Proposition 1.** *Assume the true probability vector $\boldsymbol{a}_0$ satisfies Assumption 1 and that $\Gamma = \gamma\sqrt{n/N^+}$, where $N^+$ denotes the number of condition-positive observations in the training data and $\gamma \ge 1$. Then, with probability at least $1 - m\alpha - \exp\left(-\frac{n}{2}(\gamma - 1)^2\right)$, we have*

$$\boldsymbol{a}_0^\top \boldsymbol{x} \ge \min_{a \in \mathcal{U}_{\hat{a}}} \boldsymbol{a}^\top \boldsymbol{x}.$$

A proof of Proposition 1 is included in Section EC3 in the electronic companion. Put simply, this proposition states that the probability of out-of-sample disappointment (i.e., the probability that the actual sensitivity of rule $\boldsymbol{x}$, $\boldsymbol{a}_0^\top \boldsymbol{x}$ is less than the conservative estimate $\min_{a \in \mathcal{U}_{\hat{a}}} \boldsymbol{a}^\top \boldsymbol{x}$)

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is bounded above by the probability ($\leq m\alpha$) that the sensitivity of at least one test included in rule $\boldsymbol{x}$ falls outside of its $(1-\alpha)$ confidence interval plus a term that exponentially decreases in $\Gamma^2 \propto \gamma^2$. For the latter term to remain constant, we need $\gamma$ to scale as $1/\sqrt{n}$ (i.e., $\Gamma$ to scale as $1/\sqrt{N^+}$). Such probabilistic guarantees are pervasive in robust optimization (see Bertsimas et al. 2021 and references therein). For $\Gamma = \gamma\sqrt{n/N^+}$, Bertsimas et al. (2021, corollary 3) leads to an exponential term $\exp\left(-\frac{1}{2nN^+}\gamma^2\right)$, which is worse than the exponential term in Proposition 1 yet holds without any restriction on $\gamma$.

Finally, using strong duality, we can convert the inner minimization problem in Problem (2) into one of maximization and reformulate (2) as a convex discrete optimization problem, solvable by branch and bound, as stated in Lemma 2 (the proof is included in Section EC3 in the electronic companion).

**Lemma 2.** *Problem (2) is equivalent to a mixed-integer linear optimization problem.*

In practice, we apply a similar approach to $\hat{\boldsymbol{b}}$. By considering confidence intervals on the specificity of each test,[3] $[\hat{q}_k - \hat{\theta}_k, \hat{q}_k + \hat{\theta}_k]$, we construct an analogous uncertainty set

$$\mathcal{U}_{\hat{b}} = \left\{ \boldsymbol{b} \geq 0 : \boldsymbol{e}^\top \boldsymbol{b} = 1;\ \sum_{s \in \mathcal{S}_m} |b_s - \hat{b}_s| \leq \Gamma;\ \left| \sum_{s: s_k = 0} b_s - \hat{q}_k \right| \leq \hat{\theta}_k,\ \forall k \right\},$$

and we consider a robust knapsack problem with uncertainty on both $\boldsymbol{a}$ and $\boldsymbol{b}$:

$$\max_{x \in \mathcal{X}} \left\{ \min_{a \in \mathcal{U}_{\hat{a}}} \boldsymbol{a}^\top \boldsymbol{x} \right\} \text{ s.t. } \boldsymbol{b}^\top \boldsymbol{x} \leq 1 - t_0,\ \forall \boldsymbol{b} \in \mathcal{U}_{\hat{b}}. \tag{3}$$

### 3.4. Accounting for Partial Observations

Next, we address another important source of uncertainty: missing data. Indeed, the ability to accommodate missing data is crucial for practical impact because real data routinely contain missing entries. In particular, in most studies that compare the performance of tests, all tests cannot be conducted on all individuals because of experimental failure or a limited volume of each biological sample (see Section 4.1). Hence, a complete-case analysis would lead to different sample sizes for every subset of tests, making apples-to-apples comparisons of sensitivity/specificity values difficult.

In the absence of missing data, we know all the test results for all individuals (i.e., we can associate each individual with a single outcome sequence $\boldsymbol{s} \in \mathcal{S}_m$). We have access to a matrix $\boldsymbol{Z} \in \{0,1\}^{N^+ \times n}$, where $Z_{i,s} = 1$ if individual $i$ obtains sequence $\boldsymbol{s}$. From $\boldsymbol{Z}$, we can compute the in-sample sensitivity estimates $\hat{\boldsymbol{a}} = \frac{1}{N^+}\boldsymbol{Z}^\top \boldsymbol{e}$ or equivalently,

$$\hat{a}_s = \frac{1}{N^+} \sum_{i=1}^{N^+} Z_{i,s}.$$

However, some individuals might have taken fewer than $m$ tests. We cannot associate such individuals with a single outcome sequence $\boldsymbol{s}$. Instead, for each individual $i$, we consider the subset $\mathcal{S}^{(i)}$ of all outcome sequences that are consistent with the observed test results. If individual $i$ took all the tests, $\mathcal{S}^{(i)}$ is a singleton. Formally, let $\boldsymbol{s}^{(i)} \in \{0, 1, \texttt{NA}\}^m$ denote the observed sequence of results for individual $i$. $s_k^{(i)} = 1$ (0) indicates that test $k$ is positive (negative), whereas $s_k^{(i)} = \texttt{NA}$ indicates that the result of test $k$ is unavailable. Accordingly, $\mathcal{S}^{(i)}$ is defined as $\mathcal{S}^{(i)} = \{\boldsymbol{s} \in \mathcal{S}_m : \forall k, s_k^{(i)} \neq \texttt{NA} \Rightarrow s_k^{(i)} = s_k\}$. Without imposing additional restrictions, the matrix $\boldsymbol{Z}$ belongs to the set $\mathcal{Z}^+$ defined as:

$$\left\{ \boldsymbol{Z} \in [0,1]^{N^+ \times n} : \sum_{s \in \mathcal{S}^{(i)}} Z_{i,s} = 1,\ \forall i;\ Z_{i,s} = 0,\ \forall \boldsymbol{s} \notin \mathcal{S}^{(i)} \right\}.$$

Without loss of generality, we relax the constraint $Z_{i,s} \in \{0,1\}$ to $Z_{i,s} \in [0,1]$ and interpret $Z_{i,s}$ as the likelihood that individual $i$ gets sequence $\boldsymbol{s}$.

Because $\boldsymbol{Z}$ is no longer unique, neither is $\hat{\boldsymbol{a}}$. A standard imputation technique would select one $\boldsymbol{Z} \in \mathcal{Z}^+$ (e.g., by likelihood maximization), compute $\hat{\boldsymbol{a}} = \frac{1}{N^+}\boldsymbol{Z}^\top \boldsymbol{e}$, and solve (2) for this value of $\hat{\boldsymbol{a}}$. Instead, we consider all the possible values for $\boldsymbol{Z}$ and integrate the uncertainty in $\boldsymbol{Z}$ into our robust formulation by replacing the uncertainty set in (2) by

$$\tilde{\mathcal{U}}_a = \left\{ \boldsymbol{a} : \exists \boldsymbol{Z} \in \mathcal{Z}^+ \text{ s.t. } \boldsymbol{a} \in \mathcal{U}_{\hat{a}} \text{ with } \hat{\boldsymbol{a}} = \frac{1}{N^+}\boldsymbol{Z}^\top \boldsymbol{e} \right\}.$$

The resulting problem can be formulated as a linear mixed-integer optimization problem similar to Problem (2) (see Lemma EC1 in Section EC3 in the electronic companion). We focused our exposition on the condition-positive individuals and the estimation of $\boldsymbol{a_0}$, but the same logic applies for $\boldsymbol{b_0}$. We will demonstrate the benefit of this robust imputation technique over maximum-likelihood imputation via numerical experiments.

## 4. Data and Implementation

In this section, we introduce our data and describe the application of our methods from Section 3 in the context of antibody and antigen testing.

### 4.1. Data

We apply our methodology to two data sets, provided to us by researchers on the National COVID Scientific Advisory Panel (NCSAP) in the United Kingdom and from the Charite Berlin hospital in Germany (which

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**Table 1.** Individual Performance of Nine POC LFIA Devices for Antibody Detection

| Test | $(N_+, N_-)$ | Sensitivity (95% CI) | Specificity (95% CI) |
|---|---|---|---|
| 1 | (33, 60) | 0.55 (0.47, 0.61) | 1.00 (0.93, 1.00) |
| 2 | (38, 91) | 0.61 (0.53, 0.66) | 0.99 (0.94, 1.00) |
| 3 | (33, 60) | 0.64 (0.55, 0.70) | 0.97 (0.89, 0.98) |
| 4 | (38, 60) | 0.66 (0.58, 0.71) | 0.98 (0.91, 1.00) |
| 5 | (31, 60) | 0.61 (0.52, 0.68) | 0.97 (0.89, 0.98) |
| 6 | (31, 60) | 0.65 (0.55, 0.71) | 0.98 (0.91, 1.00) |
| 7 | (33, 60) | 0.70 (0.6, 0.75) | 0.95 (0.88, 0.97) |
| 8 | (32, 60) | 0.56 (0.48, 0.63) | 1.00 (0.93, 1.00) |
| 9 | (40, 142) | 0.55 (0.48, 0.61) | 0.97 (0.94, 0.98) |

*Note.* The 95% CIs are obtained using the Wilson score method with continuity correction.

have informed the UK Government's and the World Health Organization's recommendations on COVID-19 testing, respectively). The data sets record the results of a head-to-head comparison of the diagnostic accuracy of POC tests for COVID-19 antibody and antigen detection. Note that these studies use stored samples of COVID-positive and -negative individuals (as most studies in the field do). As a result, the lack of accuracy of any given test can most likely be attributed to a limitation in the testing technology rather than the administration of the test.

**4.1.1. Antibody Tests.** The Wellcome Trust funded a performance review of nine POC lateral flow immunoassay (LFIA) devices by the NCSAP (Adams et al. 2020). In particular, their point estimates for the sensitivity (specificity) of LFIA devices range from 0.55 to 0.70 (from 0.95 to 1.00) versus the benchmark laboratory test, which led them to conclude that "currently available commercial LFIA devices do not perform sufficiently well for individual patient applications." We report the performance of the individual LFIAs in Table 1. The 95% CIs are obtained using the Wilson score method with continuity correction (Newcombe 1998).[4] Observe that the authors could not apply all tests to all samples because of the limited number of LFIA devices made available to them.

**4.1.2. Antigen Tests.** Corman et al. (2021) compared the performance of seven antigen tests. They all target the same viral protein, namely the nucleocapsid protein. Table 2 reports 95% confidence intervals for the specificity and sensitivity of each test using the same method. As noted by the authors, their experimental setting deviates from manufacturers' instructions (they conduct a predilution step for condition-positive samples). So, the measured sensitivities might be lower than in practice. Given the poor sensitivity of test II (0.09) on 45 positive samples with high viral load, Corman et al. (2021) decided not to study its sensitivity further and not to apply it on the remaining samples. To avoid any selection bias in our analysis, we will exclude test II from our study.

**Table 2.** Individual Performance of Seven Antigen Tests

| Test | $(N^+, N^-)$ | Sensitivity (95% CI) | Specificity (95% CI) |
|---|---|---|---|
| I | (105, 135) | 0.71 (0.68, 0.73) | 0.99 (0.96, 1.00) |
| II | (45, 135) | 0.09 (0.08, 0.16) | 1.00 (0.97, 1.00) |
| III | (105, 135) | 0.85 (0.81, 0.86) | 0.89 (0.86, 0.90) |
| IV | (105, 135) | 0.28 (0.26, 0.31) | 1.00 (0.97, 1.00) |
| V | (105, 135) | 0.76 (0.73, 0.78) | 0.95 (0.92, 0.96) |
| VI | (105, 135) | 0.34 (0.33, 0.37) | 0.99 (0.96, 1.00) |
| VII | (115, 135) | 0.70 (0.67, 0.72) | 0.99 (0.95, 0.99) |

*Note.* The 95% CIs are obtained using the Wilson score method with continuity correction.

The summary statistics on our data, reported in Table 3, motivate two useful observations. First, the sizes of both data sets are limited, so the estimates for the diagnostic accuracy of the individual tests are subject to finite-sample estimation error. This motivates our robust formulation, which accounts for such uncertainty (applied in Section 5.3). Second, both data sets suffer from missing data. Indeed, in both cases the authors were unable to apply all the tests to all samples because of limited testing kit availability, limited volume per sample, or experimental failure. A naïve complete case analysis would reduce the already limited sample size (and increase the estimation error). This motivates our robust formulation, which additionally accounts for missing data (evaluated in Section 5.4).

Finally, we note that our data contain no precise information on the cost of each of these different tests. As mentioned earlier, data on test costs are scarce (Xiao and Rathi 2021), and costs depend on many factors and vary widely. Because each data set compares tests that are similar to manufacture and administer (e.g., LFIA

**Table 3.** Characteristics of the Diagnostic Testing Data Sets Used in Our Numerical Experiments

| Data set | $N^+$ | $N^-$ | $m$ | Complete cases | Missing entries, % |
|---|---|---|---|---|---|
| Antibody tests (Adams et al. 2020) | 40 | 142 | 9 | 89 | 41 |
| Antigen tests (Corman et al. 2021) | 138 | 135 | 6 | 207 | 11 |

*Note.* $N^+$ ($N^-$) denotes the number of condition-positive (condition-negative) samples, and $m$ is the number of tests.

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devices), in the absence of more precise information, we will work under the assumption that all tests have comparable (yet unknown) cost. Accordingly, we can set $c_k = 1$ in the budget constraint defining $\mathcal{X}$ and in our analysis, identify the overall cost of a combination testing policy with the number of tests involved, $K$. Nonetheless, as previously mentioned, combining a few POC tests would remain cheaper than conducting a laboratory-based test.

### 4.2. Motivation for Combining COVID-19 Tests

Intuitively, combining tests can be beneficial for systematic or stochastic reasons. First, each test might use a different underlying technology, hence providing its unique view on the patient's condition. Second, a test result can be viewed as a stochastic signal and combination testing as a denoising procedure. We illustrate these two mechanisms on two stylized examples in the appendix. In this section, we provide examples of different underlying technologies or sources of randomness in the case of COVID-19 detection.

First, consider the systematic mechanism. Technological differences across tests may allow one test to detect a target protein for a given patient, even if another test does not. Antigen tests detect the presence of virus proteins. For COVID-19, tests usually target the nucleocapsid protein or the spike protein by mixing the patient's sample with synthetic antibodies that have high affinity toward one subunit of the protein of interest. Because these proteins might be present in infected patients at different concentrations, for instance, and because the synthetic antibodies used might be constructed differently, the tests might give discordant results. For antibody tests, two types of antibodies are typically used (Immunoglobulin M (IgM) and Immunoglobulin G (IgG)), and manufacturers implement different detection strategies. Combining multiple tests can therefore result in improved accuracy by leveraging the different testing technologies. Similarly, Blacksell et al. (2011) and Veyrenche et al. (2021) consider detecting current infections using one antigen and one IgM antibody test.

Second, consider the stochastic mechanism. Each test result can be viewed as the realization of a random variable, and conducting multiple tests can reduce the noise associated with each application of a test. At the patient level, this stochastic view is supported by the presence of inherent noise stemming from how the test is administered and analyzed. For example, even applying multiple tests that rely on the same technology on the same individual might not consistently give the same result. Combining the results from multiple tests can have higher accuracy because it reduces noise, in the same way that diversification reduces the risk of a portfolio. In addition, some individual characteristics (e.g., viral load, time since infection, presence of symptoms) can impact test results and accuracy. At a patient level, they are constant, but they contribute to randomness (and correlation) in test results at the population level because they are unobserved and heterogeneous across patients.

Our methodology aims to leverage both of the mechanisms described in a disciplined way to achieve higher classification accuracy. In an attempt to disentangle and quantify the extent to which systematic or stochastic mechanisms are prevalent, we compute and report (in Section EC5.1 in the electronic companion) two metrics for each pair of tests on condition-negative and condition-positive samples separately. To measure the extent of agreement among tests, we use the probability of pairwise agreement. To measure stochastic association between tests, we report their pairwise correlations. These descriptive statistics are included in Section EC5.1 in the electronic companion; Tables EC1 and EC3 in the electronic companion describe the antibody test data, whereas Tables EC2 and EC4 in the electronic companion describe the antigen test data.

We observe that pairwise correlations are close to zero on condition-negative individuals for both data sets. This suggests that discrepancy in test results is mostly because of independent noise. On the contrary, for condition-positive samples, agreement probabilities are low, demonstrating that tests give discordant results more often on condition-positive samples than on condition-negative samples. Furthermore, correlations are higher as well, suggesting that such discordance in these test results cannot be solely explained by independent random noise and that there might be underlying factors (e.g., time since infection for antibody tests or viral load for antigen tests) or systematic technological differences that correlate test results on condition-positive samples.

### 4.3. Application of Methods to COVID-19 Data Sets

In our implementation of the robust model of Section 3.3, we will consider uncertainty sets for $\boldsymbol{a}$ of the form

$$\mathcal{U}_{\hat{a}}(\Gamma) = \left\{ \boldsymbol{a} \geq 0 \;:\; \boldsymbol{e}^\top \boldsymbol{a} = 1; \right.$$
$$\left. \sum_s |a_s - \hat{a}_s| \leq \Gamma; \; \left| \sum_{s:s_k=1} a_s - \hat{p}_k \right| \leq \hat{\rho}_k, \; \forall k \right\},$$

where $\hat{p}_k$ and $\hat{\rho}_k$ are set equal to the middle and half-width of the CIs for the sensitivity of test $k$ (Tables 1 and 2) and $\Gamma \geq 0$ is a hyperparameter, which we calibrate using fivefold crossvalidation (CV). A detailed description of the CV procedure is provided in Section EC4 in the electronic companion. We similarly construct uncertainty sets $\mathcal{U}_{\hat{b}}(\Gamma)$ for $\boldsymbol{b}$.

We make two notes about the presentation of our results. First, diagnostic accuracy is generally reported

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in terms of sensitivity and specificity. Improvements in one of these metrics (e.g., sensitivity) only make sense if the other (e.g., specificity) is capped at a given threshold. For instance, we report achievable improvement in sensitivity given varying specificity targets in Section 5.5. However, setting a relevant value for this threshold will depend on the use case. Accordingly, we use the AUC as an accuracy metric in the CV procedure (Algorithm 2) and present our main results in terms of AUC because it captures in a single number the sensitivity/specificity trade-off in a threshold-free manner.[5] AUC corresponds to the probability of correctly ranking a condition-positive individual higher than a condition-negative one, which is especially relevant when screening provides access to a scarce and expensive resource. Still, we would like to emphasize that our method is agnostic to the metric being used. Other measures of accuracy can be used, both for crossvalidation and reporting of the results, if deemed more relevant to the application at hand.

Second, when selecting $K$ of $m$ tests to combine, we solve a knapsack problem—(1) or (3)—for varying specificity thresholds $t_0$, hence generating a Pareto-dominating ROC curve. Each point on that curve corresponds to a policy, namely $K$ tests and a rule to combine their results. Accordingly, the AUC of the Pareto-dominating ROC curve does not correspond to the AUC of a unique $K$-test combination (the $\boldsymbol{y}$ variable) but rather, captures the overall predictive power of the available tests when combined, representing the state of technology. For $K$ fixed, among all the subsets of $K$ tests that lie on this Pareto-dominating ROC curve, we conservatively focus our attention on the best (i.e., AUC-maximizing) one. This allows a fairer comparison with individual tests and emphasizes an application of our method that is simpler to implement in practice. Our results for the complete Pareto-dominating ROC curve (for the optimal combinations of $K$ tests at any specificity threshold) are discussed in Section EC5 in the electronic companion.

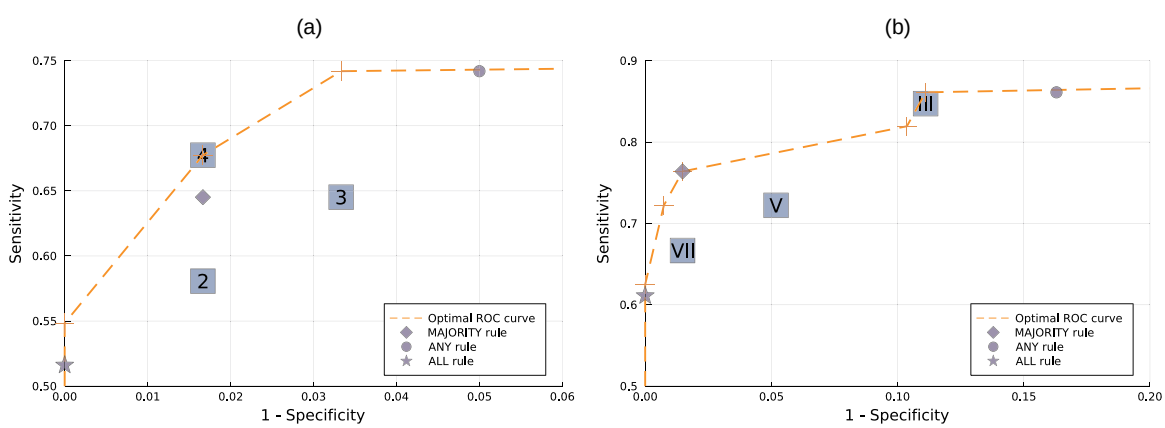
## 5. Results

We demonstrate the benefits of implementing our combination testing policies using data on COVID-19 antibody and antigen tests, described in Section 4.1. We first illustrate the dominating ROC curve for a given subset of tests in Section 5.1. We then discuss the improvement in accuracy achievable by optimally selecting the tests to include in a combination policy in Section 5.2 and the out-of-sample performance of our robust formulations that address estimation error and missing data in Sections 5.3 and 5.4, respectively. We then compare the performance of a combination of tests selected using our methodology with the performance of heuristics previously proposed in the literature in Section 5.5. Finally, we provide an illustration of how a policy maker would apply our methodology to obtain a specific combination testing policy in Section 5.6. In addition, Section EC6 in the electronic companion contains results on synthetic data, exploring the impact of the sample size and the proportion of missing entries on our results.

### 5.1. Combining a Given Set of Tests

For illustration, we first restrict our attention to three arbitrarily chosen tests from each data set. We obtain all Pareto-optimal combination rules for these tests, plotted as the break points in the dashed (orange) ROC frontier in Figure 1, by solving the knapsack Problem (1) for different values of $t_0$. As discussed in Section 3.1, the frontier can be computed by sorting the eight possible outcome sequences by their $\hat{a}_s/\hat{b}_s$ values. For now, we consider only complete cases. We observe from Figure 1 that the dashed ROC frontier dominates, at least weakly, the performance of all of the three individual tests, thus illustrating the benefit from combining tests. We also

**Figure 1.** (Color online) The Trade-Off Between Specificity and Sensitivity (ROC Curve) for Three Individual Tests and All Their Potential Combinations




*Notes.* (a) Antibody test data. (b) Antigen test data.

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observe that the standard heuristics discussed in Section 2 are not guaranteed to lie on the frontier and that our methodology identifies points on the ROC frontier that the heuristics cannot (Section 5.5 and Section EC5.2 in the electronic companion contain a more extensive comparison with these heuristics).

Although Figure 1 demonstrates the performance of our methodology for a given set of tests, a key part of the practical challenge is to select which tests to combine, especially given the proliferation of rapid tests. For example, our antibody (antigen) data set comprises nine (six) different tests, leading to 36 (15) possible two-test combinations and 84 (20) three-test combinations. In the following three sections, we efficiently search for the best combination of three antibody (or antigen) tests by solving our extended formulation with additional binary variables $\boldsymbol{y}$, and we illustrate how robustness mitigates the issues of overfitting and missing data.

## 5.2. Selecting a Combination of Tests

We next illustrate the improvement in diagnostic accuracy from optimally selecting the tests to be included in the combination testing policy. We consider all available tests and solve (3) with $\Gamma = 0$ (i.e., obtaining only in-sample estimates of $\hat{\boldsymbol{a}}$ and $\hat{\boldsymbol{b}}$) on all of the complete cases and construct the ROC curve for the individual tests as well as for the Pareto-optimal combinations of two and three tests. Figure 2 displays both the performance of the individual tests (labeled by their index number) as well as the ROC frontiers of combinations of two (the orange dashed line with "×" break points) and three (the green dashed line with "+" break points) tests. We observe that the ROC frontiers for combining either two or three tests lie above the convex hull of the index tests.

On the antibody test data, although the best-performing test (test 7) has an in-sample AUC of 0.837, the set of Pareto-optimal three-test combinations achieves 0.894 AUC (+7%). It also identifies two three-test combinations—(6, 7, 8) and (1, 6, 8)—with AUCs of 0.862 and 0.888, respectively. Similarly, on the antigen test data, we observe a 5% increase in AUC between the best-performing test (test III, AUC = 0.868) and the Pareto-optimal set of three-test combinations (AUC = 0.921). Among the three-test combinations on the Pareto-dominating ROC curve, (I, III, V) is the most accurate one (AUC = 0.878).

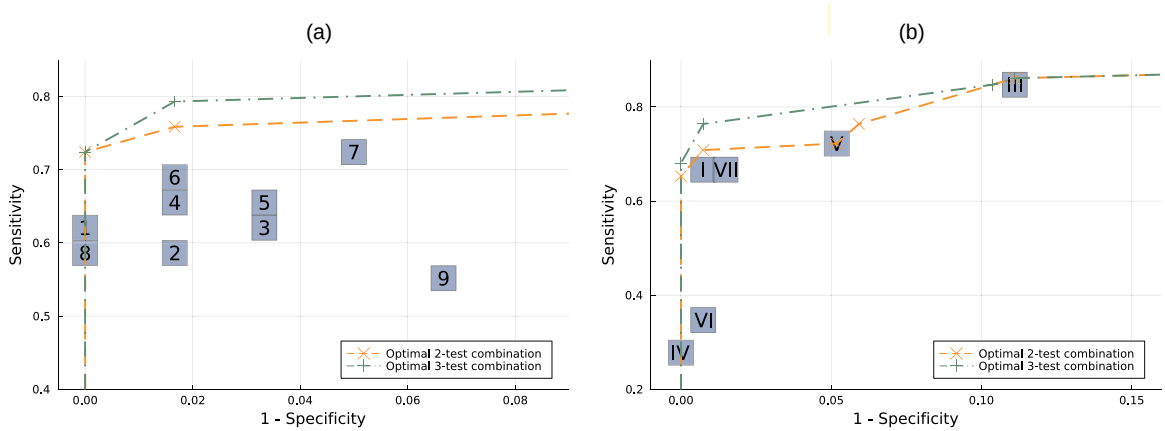
## 5.3. Robustness to Parameter Estimation

Because of finite-sample estimation error, our optimization model might find combination policies that improve accuracy on the training data but do not generalize to the entire population.

To illustrate the value of robust optimization in mitigating this concern, we repeatedly split the data into training and validation data sets using a 70%/30% split (the robustness of the results with respect to the splitting proportion is presented in Section EC5.2 in the electronic companion (Figure EC1 in the electronic companion) using stratified sampling to keep the prevalence constant between the two data sets). For each training data set, we apply the following methodology for each $K$; we estimate the sensitivity and specificity of each test and their 95% CIs, construct uncertainty sets of the form $\tilde{\mathcal{U}}_{\hat{a}}(\Gamma)$ ($\tilde{\mathcal{U}}_{\hat{b}}(\Gamma)$) for $\boldsymbol{a}$ ($\boldsymbol{b}$), and compute the Pareto-dominating ROC curve. From this curve, we identify the most accurate $K$-test combination on the training data and evaluate its performance on the validation data. We repeat this procedure over 50 random splits.

Figure 3 compares three performance curves. First, it reports in- and out-of-sample performance of the nonrobust solution ($\Gamma = 0$) with missing data imputed using ML imputation (Gelman and Hill 2006, chapter 25). On both data sets, we observe that the out-of-sample

**Figure 2.** (Color online) The Trade-Off Between Specificity and Sensitivity (ROC Curve) for All the Individual Tests and the Optimal Combinations of Two (Dashed Lines) and Three Tests (Dash-dotted Lines)




*Notes.* (a) Antibody test data. (b) Antigen test data.

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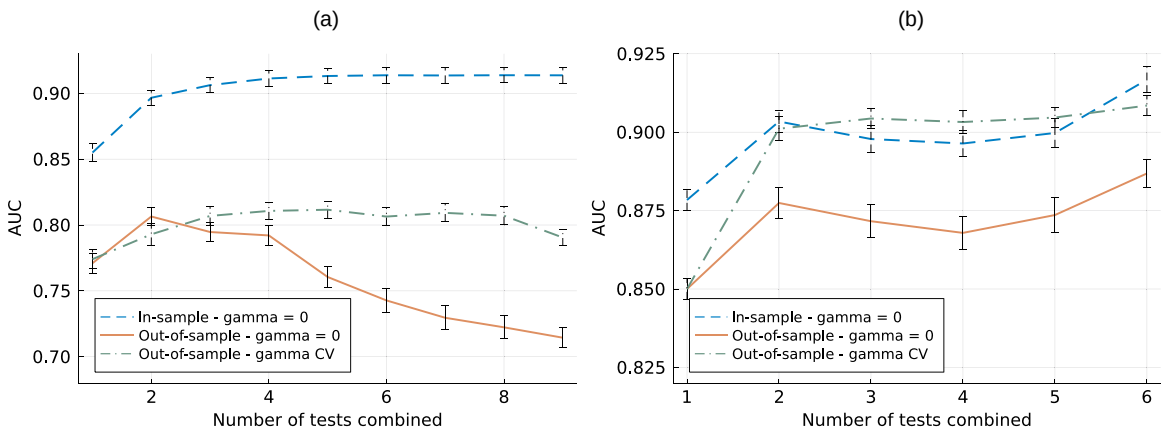
**Figure 3.** (Color online) Diagnostic Accuracy (AUC) of the Best $K$-Test Combination as a Function of the Number of Tests Combined ($K$)




*Notes.* Error bars (in all figures) correspond to ±1 standard error. (a) Antibody test data. (b) Antigen test data.

AUC (solid orange line) is significantly lower than its in-sample value (dashed blue line). This discrepancy is starker on the antibody test data; although in-sample AUC unambiguously increases as more tests are used, out-of-sample performance deteriorates when using three tests or more.

Second, the dashed-dotted green line in Figure 3 displays the out-of-sample AUC of a robust policy where the value of $\Gamma$ is calibrated using a fivefold CV procedure (see Section EC4 in the electronic companion for pseudocode) on the training data set and where missing data are imputed adversarially. The figure illustrates that adding robustness can improve out-of-sample performance. In both cases, the robust models exhibit comparable or stronger out-of-sample performance than the naïve approach. For instance, our method identifies three-test combinations with an average AUC of 0.802 on the antibody test data (+4% compared with 0.774 for a single test) and of 0.902 on the antigen test data (+6%). For moderate values of $K$ ($K \leq 4$), which are probably the most relevant values for policy implications, the benefit from robustness is mild on the antibody test data but stark on the antigen test data. In the electronic companion, we report similar yet smaller gains for the entire Pareto-dominating curve instead of the best $K$-test combination only (in Figure EC2 in the electronic companion).

The behavior of the method displayed on Figure 3 is notably different on the antibody and antigen test data. On the antibody test data, the nonrobust optimization formulation provides good solutions for up to three tests, but its out-of-sample performance deteriorates for larger values of $K$. The robust formulation does not significantly improve on the nonrobust formulation for small values of $K$ but stabilizes the out-of-sample AUC for $K \geq 4$. On the contrary, on the antigen test data, the accuracy of the nonrobust formulation generally increases with $K$, and the robust formulation provides substantial improvement for all $K \geq 2$. One explanation could be the difference in sample sizes. The total number of observations (and of condition-positive samples) in the antigen data is more than twice (three times) as high as for the antibody data, and there are fewer missing data. An alternative explanation could be that among the tests in our data, the antibody tests might rely on more similar technologies than the antigen tests, so combining tests helps mostly because it reduces noise, not because we gain new information. To support this intuition, we decompose the results for antigen tests on high- and low-viral load samples separately in Section EC5.3 in the electronic companion. In summary, our (non-)robust combination policies display a similar behavior on the high-viral load samples to what we observe for the antibody test data. This observation corroborates the intuition that when various technologies are available and tests are systematically discordant (as in the low-viral load antigen tests data), our method is able to find particular classification rules that leverage these different signals, whereas when technologies are more mature and similar, combination of tests helps mainly as a denoising procedure.

### 5.4. Robustness to Missing Data

The robust optimization approach we consider in the previous section differs from the baseline approach in two ways. First, it imputes the missing values adversarially instead of using maximum likelihood. Second, it accounts for noisy estimates by considering uncertainty sets around the empirical values $\hat{\boldsymbol{a}}$ and $\hat{\boldsymbol{b}}$ (with $\Gamma$ cross-validated instead of $\Gamma = 0$). Both ingredients provide some form of robustness and could explain the observed improvement. To isolate the relative benefit of each of them, Figure 4 compares the out-of-sample performance of two models nonrobust to noisy estimates ($\Gamma = 0$): one where missing entries are imputed with their most likely

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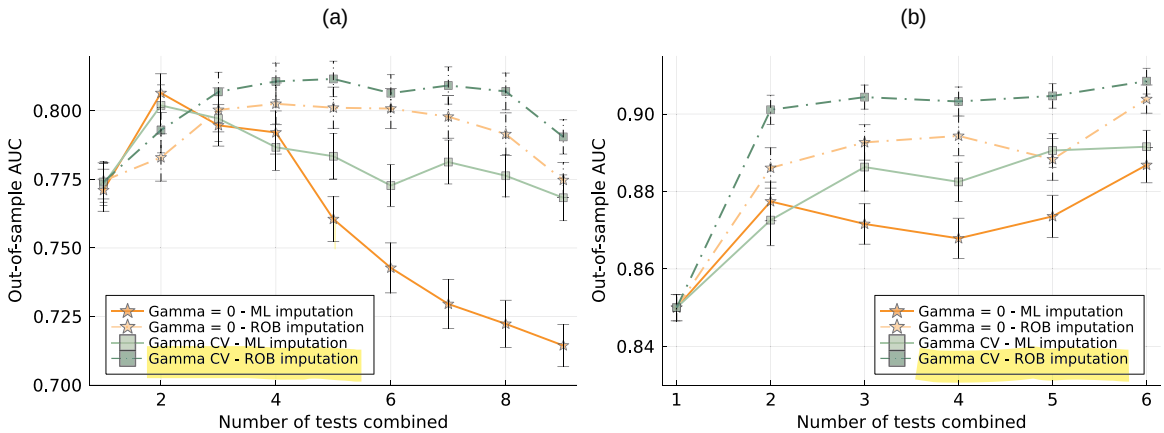
**Figure 4.** (Color online) Out-of-Sample Accuracy (AUC) as a Function of the Imputation Method for Missing Data




*Notes.* The orange solid lines and the dashed green lines are the same as in Figure 3. (a) Antibody test data. (b) Antigen test data.

values as in Figure 3 (orange solid line in Figure 4) and another where they are imputed adversarially (orange dashed-dotted line in Figure 4).

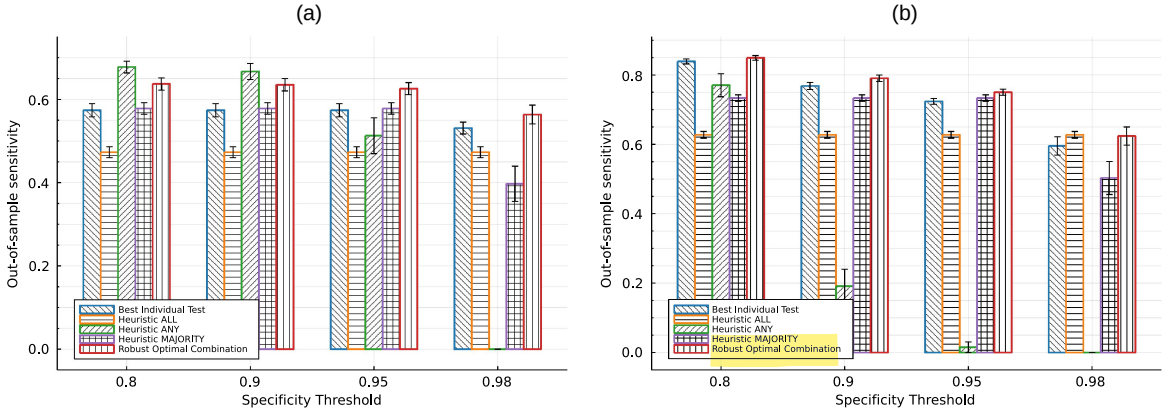
We observe that robust imputation (labelled ROB in Figure 4) provides a substantial improvement over ML imputation, especially as $K$ increases. For comparison purposes, we also report the performance of both imputation methods with additional robustness to noisy estimates ($\Gamma$ CV) in green. In summary, irrespective of the value of $\Gamma$ (zero or CV), adversarial imputation improves out-of-sample predictive power over ML imputation (dashed-dotted versus solid lines). Irrespective of the imputation method, adding robustness ($\Gamma$ CV) helps (green versus orange). The robust formulation with adversarial imputation provides the best performance.

## 5.5. Comparison with Existing Heuristics

As discussed in Sections 1 and 2, the prior literature has proposed various heuristics for interpreting the outcomes of multiple tests performed to detect the same condition. In particular, the `any` (if any test is positive), `all` (if all tests are positive), and `majority` (if the majority of tests are positive) heuristics have been proposed.

We first compare the performance of the set of Pareto-optimal three-test combinations with that of a single test and of the three aforementioned heuristics applied on the three individually best-performing tests for each data set. On the training data, for each specificity target, we select the best test (or three-test combination) in terms of sensitivity and compute its out-of-sample sensitivity on the validation set. We focus on three-test combinations because Figure 3 demonstrates no additional benefits from adding more tests to a combination. For four specificity targets, Figure 5 reports the average out-of-sample sensitivity achievable by a single test, the heuristics, and our method. We observe that using our methodology to identify the best combination of tests and the appropriate classification rule to use at each threshold results in a significant improvement in sensitivity for almost all specificity thresholds. In particular, if

**Figure 5.** (Color online) Out-of-Sample Sensitivity Achieved by Combining Three Tests for Varying Specificity Thresholds




*Notes.* The performance of the heuristics is obtained by identifying the three best-performing individual tests for antibody (antigen) detection and applying the three heuristics. (a) Antibody test data—three test combinations. (b) Antigen test data—three test combinations.

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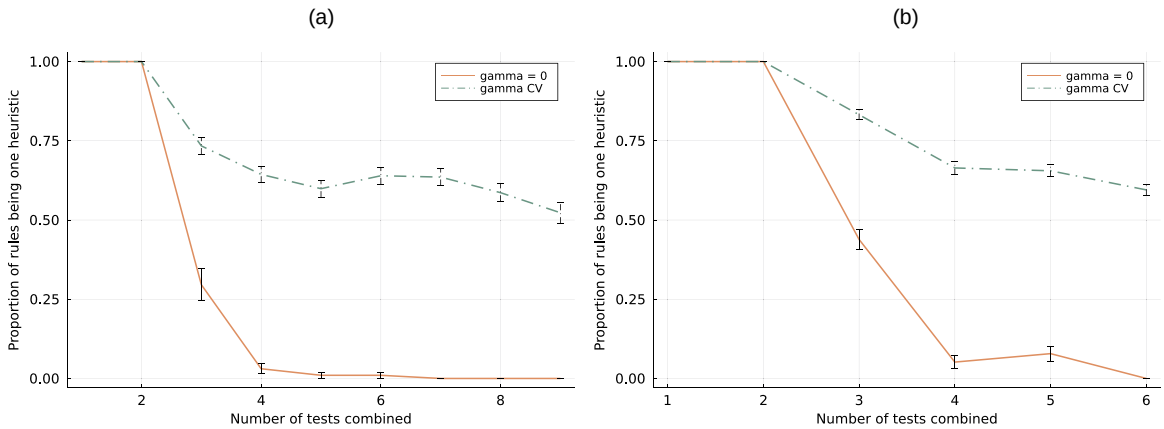
**Figure 6.** (Color online) Fraction of the Pareto-Optimal $K$-Test Combinations Corresponding to One of the Heuristic Rules (`All/Any/Majority`)




*Notes.* (a) Antibody test data. (b) Antigen test data.

a policy maker requires specificity of at least 0.95, our methodology results in 8% and 2% relative improvements in sensitivity for antibody and antigen detection, respectively, compared with the best-performing heuristic (`majority` in these cases). Over the eight cases (4 specificity targets × 2 data sets), our methodology significantly ($p$-value of <0.1) outperforms the available heuristics in four cases. Our method is only dominated (with a $p$-value of <0.1) by the `any` heuristic, on the antibody test data, and for a specificity threshold of 0.8. This behavior is most likely because of our crossvalidation procedure for $\Gamma$ (see the pseudocode in Algorithm 2 in Section EC4 in the electronic companion), which optimizes for the aggregate AUC measure. Optimizing for partial AUC (or sensitivity conditional on at least 0.80 specificity) would likely lead to improved performance for our approach. Similarly, our optimal three-test combination systematically improves over using a single test, with the comparison being statistically significant in four of the eight cases.

As previously discussed, our optimization method achieves two concurrent objectives: to identify a subset of $K$ tests (as encoded by the variable $\boldsymbol{y}$) and then, to find the best classification rule to aggregate their results (as encoded by the variable $\boldsymbol{x}$). In contrast, heuristic rules solely address the issue of classification and not of test selection. We now try to disentangle the respective benefit of our method on the test selection and classification tasks separately.

In Figure 6, we consider the entire Pareto-optimal set of $K$-test combinations and count how many of them correspond to one of the aforementioned heuristics on the subset of tests they involve.[6] Obviously, for $K \leq 2$ tests, any classification rule can be seen as an `any` or `all` rule. As the number of tests involved, $K$, increases, the fraction of Pareto-optimal combinations that correspond to one of the heuristics decreases. However, we observe that our robust optimization formulation recovers heuristic rules substantially more frequently than the nonrobust version does, suggesting that the heuristics are somewhat robust classification rules and that the optimal selection of tests is a central added value of our method. Still, for three tests, around 25% (20%) of the Pareto-optimal combinations are more complex than the heuristics on the antibody (antigen) data.

To further illustrate the benefit of our optimization formulation in optimally selecting which test to combine, we solve our robust optimization Problem (3) with additional constraints that force the classification rule $\boldsymbol{x}$ to be one of the aforementioned heuristics (see the detailed formulation in Section EC1 in the electronic companion). Doing so provides the existing heuristics with a disciplined approach to identify the most powerful combination of tests while accounting for in-sample estimation error and missing data. Figure 7 replicates Figure 5, except that tests for the heuristic rules are selected optimally instead of heuristically. We observe that test selection has a first-order impact on the accuracy of a combination policy; optimally selecting the tests substantially increases the performance of all heuristic rules, up to the level of the optimal combination. For example, for a specificity of at least 0.98, the `majority` rule achieves a sensitivity of 0.56 (0.69) on the antibody (antigen) data when tests are optimally selected compared with 0.40 (0.50) when the best index tests are selected and compared with 0.56 (0.62) for the optimal combination strategy.

### 5.6. An Illustration of a Specific Policy Recommendation

Up to this point, our analysis has been focused on evaluating our methodology for uncovering robust combination testing policies, hence the emphasis on crossvalidation and out-of-sample accuracy. We now illustrate how a decision maker could apply our methodology in practice.

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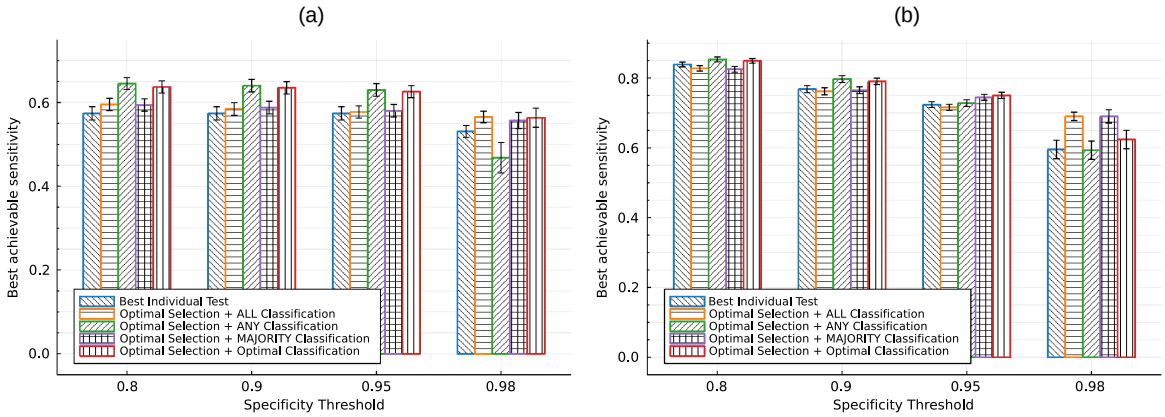
**Figure 7.** (Color online) Out-of-Sample Sensitivity Achieved by Optimally Combining Three Tests for Varying Specificity Thresholds



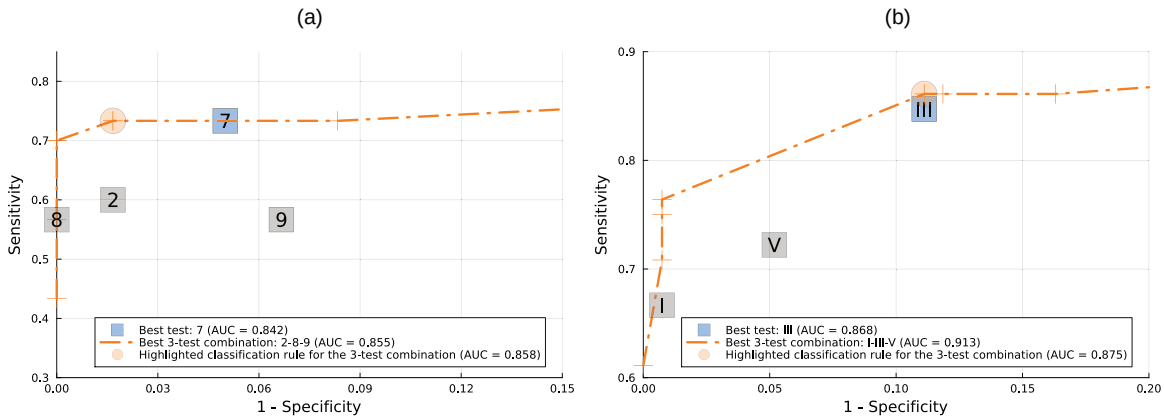

*Notes.* Tests are selected using our robust knapsack formulation. Test results are classified using the `any`, `all`, `majority`, and optimal rules. (a) Antibody test data—three test combinations. (b) Antigen test data—three test combinations.

In the simplest use case, in which the decision maker has an abundance of data, they would apply our methodology on a training data set and evaluate out-of-sample performance on a validation data set. However, in realistic use cases of our methodology, the decision maker would have access to a similar data set as our data set, which compares the performance of various POC tests on a set of samples. Those data sets tend to be smaller in size, so omitting data for training purposes can significantly affect the solutions obtained and result in a too small validation data set for a representative out-of-sample performance estimate. Therefore, for a realistic illustration of our methodology, we propose the following approach based on the notion of *stability selection*, a method widely used for variable selection in statistics (Meinshausen and Bühlmann 2010).

As a first step, we leverage our data set to identify the best single test and the best three-test combination. We randomly split the data into training (70%) and testing (30%) 50 times (the same approach as we use for crossvalidation). For each training data set, we use our methodology to identify the best-performing single test and the best-performing three-test combination—with an associated decision rule—as measured by their respective in-sample AUCs. Using stability selection, we aggregate the results by counting how frequently each single test and each three-test combination have the highest in-sample AUCs across all 50 training sets. As a second step, we evaluate the best single test and the best three-test combination—identified using the stability selection approach described—by calculating their respective AUCs on the whole data set.

Figure 8 shows the outcome of this process. We make four observations based on these results. First, for both the antibody and antigen test data sets, a combination of three tests achieves a higher AUC than the best

**Figure 8.** (Color online) ROC Curves for the Best Individual Test and the Best Three-Test Combination (Dash-dotted Lines) Identified via Stability Selection




*Notes.* (a) Antibody test data. (b) Antigen test data.

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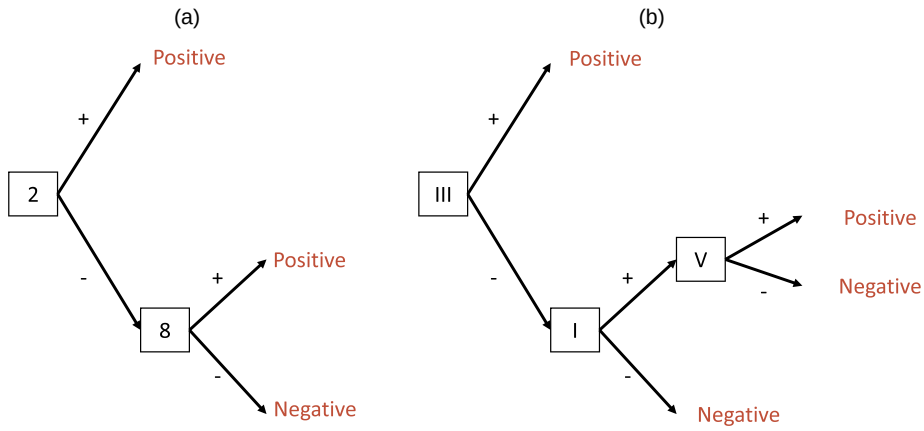
**Figure 9.** (Color online) Decision Rules Associated with the Highlighted Points in Figure 8




*Notes.* (a) Antibody test data. (b) Antigen test data.

single test. For antibody testing (Figure 8(a)), a three test combination of tests 2, 8, and 9 achieves an AUC of 0.855, whereas the best single test (test 7) achieves an AUC of 0.842, a relative improvement of 1.5%. For antigen testing (Figure 8(b)), a three-test combination of tests I, III, and V outperforms the best single test (test III) by a relative 5% in terms of AUC.

Second, an important benefit of combination testing is that the same set of (e.g., three for this illustration) tests gives the decision maker increased flexibility to emphasize either sensitivity or specificity by choosing a specific point on the ROC curve to implement. For example, using the same three antibody tests, the decision maker can achieve either a sensitivity of 0.73 (with a corresponding specificity of 0.92) or a sensitivity of 0.70 (with a corresponding specificity of 1.00). Similarly, for the selected antigen tests, the decision maker can choose between a sensitivity of 0.86 (with a corresponding specificity of 0.84) or a sensitivity of 0.61 (with a corresponding specificity of 1.00).

Third, this analysis provides an answer to the empirical question of *how much* sensitivity or specificity can be improved by combining tests in a specific application context. If the decision maker is satisfied with the sensitivity of the best-performing antibody test (test 7), specificity can be improved by 0.03 by combining tests (see the highlighted point in Figure 8(a)). On the other hand, if the decision maker is satisfied with the specificity of the best antigen test (test III), sensitivity can be only modestly improved (by 0.01) by combining three tests (see the highlighted point in Figure 8(b)).

Fourth, each point on the ROC curve for a three-test combination corresponds to a specific decision rule. If the decision maker wishes to improve antibody specificity by 0.03 (by choosing the highlighted point in Figure 8(a)), the associated decision rule is to apply the `any` rule using tests 2 and 8. In other words, even if we allow for up to three tests in the combination, this particular point on the ROC curve only requires the use of two tests. If the decision maker wishes to improve the sensitivity of antigen testing by 0.01 (by choosing the highlighted point in Figure 8(b)), the decision maker would classify a sample as positive if test III returns a positive result or if both tests I and V return a positive result. The decision rules associated with these two points are illustrated in Figure 9. Note that such compact representations of a classification rule as a test sequence can be obtained systematically using optimization, as described in Section EC2 in the electronic companion.

## 6. Conclusion and Implications

Despite advances in COVID-19 treatment and vaccination, the need for cheap, fast, and accurate testing remains. We contribute to the ongoing debate on the best use of POC devices for testing against COVID-19. We present a novel methodology, based on robust optimization, to best select and combine tests, hence providing a structured way for policy makers to quantify the benefit from combining multiple index tests and potentially break the trade-off between accuracy and cost in diagnostic testing. Our approach accounts for incomplete or noisy data, two significant practical considerations when working with limited data on emerging pathogens. We demonstrate the relevance and effectiveness of our algorithm on data sets composed of COVID-19 antibody and antigen test results from two medical institutions.

Our results indicate that, by combining multiple POC tests, widespread COVID-19 testing can be simultaneously accurate, timely, and cost-effective—and thus, could help curb transmission while reducing unnecessary isolation. In a single week of June 2021, nearly 4% of British pupils missed school because of required isolation (UK Department of Education 2021). In the United States, millions of passengers travel by air daily (Transportation Security Agency 2021). Initial guidelines imposed a negative RT-PCR result for all passengers at

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most 72 hours prior to departure, although they could be exposed and become infectious in the interim. Quick and accurate POC testing in these and other settings could reduce the costs of both false negatives and false positives.

With combination testing, a policy maker has access to a fine-grained ROC curve and can decide to emphasize either sensitivity (e.g., for vulnerable populations for which false negatives are costly) or specificity (e.g., in population-level seroprevalence studies) by selecting the appropriate classification rule. Designing combination rules that optimize for both downstream operational performance and predictive power is another interesting future direction. For example, our knapsack framework can incorporate additional operational constraints, such as limited availability of healthcare professionals to oversee or conduct testing. Overall, our method provides a practical and time-efficient method for policy makers to (re-)assess the benefit from combination testing as new tests or new variants emerge.

Future research could apply our methodology to the repeated use of the same test as well. In that setting, instead of outcome sequences, one would describe test results by the number of positive or negative outcomes only. However, data on the repeated use of a single test are, to our knowledge, not commonly published.

In general, treatment policies and decisions based on statistics are ubiquitous. However, the underlying statistical tools rarely take operational constraints into account. By formulating a foundational result in statistics, the Neyman–Pearson lemma, as an optimization problem, we are able to account for important practical considerations, such as costs, and noisy or incomplete input data. Data are scarce in many healthcare applications, so conclusions based on simple statistical approaches might overfit the data. In our case, a naïve in-sample analysis would always conclude that adding more tests to a combination would improve diagnostic accuracy. Our robust formulation builds in safeguards against policy decisions based on overfitting to limited data. As such, our paper illustrates the power of optimization as a tool to integrate statistics and operations.

Beyond the COVID-19 pandemic, our method provides a theoretically grounded and robust way to achieve higher diagnostic accuracy from combining cheap and fast predictors and could be relevant in other areas of medicine, healthcare operations, manufacturing, or service systems.

## Acknowledgments

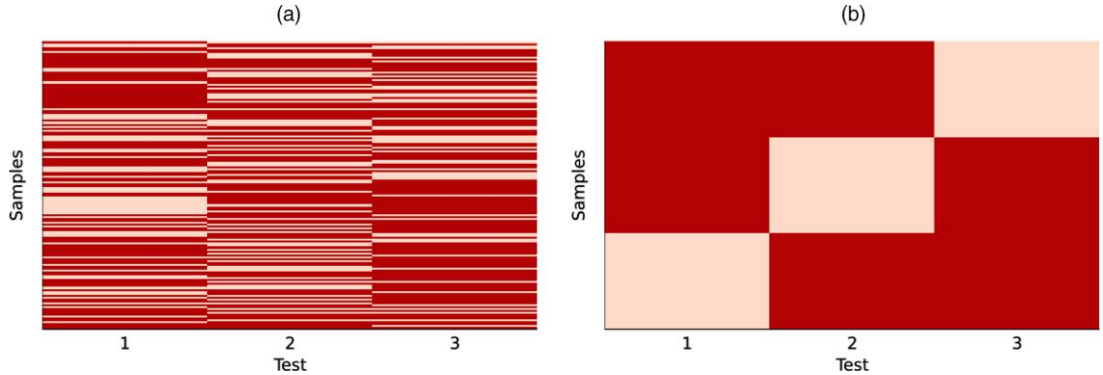
The authors thank Emily R. Adams and Victor M. Corman (the corresponding authors of Adams et al. (2020) and Corman et al. (2021), respectively) for providing access to the data on antibody and antigen testing, respectively. The authors are also grateful to Claudia Denkinger for providing useful feedback.

## Appendix. Why Combining Tests Helps: Two Stylized Examples for Intuition

Intuitively, index tests may disagree for distinct reasons. On the one hand, they may not agree because of the different underlying technology they use (e.g., the target protein or the reagents applied) and because they have different underlying capability to identify the same target protein.[7] In this case, test outcomes would have (a potentially negative) nonzero correlation. On the other hand, they may disagree because of stochastic noise associated with any testing process, in which case their results might disagree but have zero correlation.

Consequently, our data-driven approach for combining tests can improve diagnostic accuracy through two mechanisms corresponding to the two types of potential reasons for disagreement. First is a systematic mechanism because our approach leverages systematic differences in the technology of tests. Second is a stochastic mechanism because combining tests can reduce the inherent noise associated with testing. To illustrate these two mechanisms, we consider two extreme situations depicted in Figure A.1. For illustration, we focus our attention on condition-positive samples. In both situations, the three index tests (depicted on the $x$ axis) each have a sensitivity of two thirds. Each horizontal band in the figure

**Figure A.1.** (Color online) Test Results for Condition-Positive Patients (Red Is Positive) in Two Stylized Situations




*Notes.* (a) Uncorrelated tests. (b) Systematically discordant test.

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(corresponding to a specific $y$ value) represents the results on a particular sample from the three tests. Red indicates a positive test result, and pink indicates a negative test result.

• First consider the case in which there are no systematic or technological differences across tests, just stochastic noise in the test outcomes. We model the three tests as independent Bernoulli random variables, each with a sensitivity of two thirds (panel (a) in Figure A.1). Hence, two particular tests (e.g., tests 1 and 3) provide the same result with probability $(1/3)^2 + (2/3)^2 = 5/9\,(> 1/2)$ (i.e., they agree on more than half the cases but have zero correlation by construction).

• Second, consider a case in which there are clear systematic differences across tests. Each test is completely ineffective on a (different) third of the population and perfectly accurate on the remaining two thirds (panel (b) in Figure A.1).[8] In this case, two particular tests (e.g., tests 1 and 3) provide the same results with probability $1/3\,(< 1/2)$, but the results are correlated (pairwise correlation between any two tests is $-1/2$).

A combination testing policy requires (a) choosing the tests to combine and (b) deciding how to interpret the outcome sequence of the selected tests. For illustration, we assume that we wish to combine all three tests and that the decision rule we apply is the `majority` rule.

• For the first case (uncorrelated but noisy test outcomes), applying the `majority` rule yields a sensitivity of $20/27 > 2/3$. In this situation, each test corresponds to a noisy observation of the quantity of interest, and combining (in the case of the `majority` rule, "averaging") test results achieves higher predictive power because of denoising.

• For the second case (technologically different tests), applying the `majority` rule yields a sensitivity of one. In this situation, there is no stochasticity in the testing procedure per se, but each test sees only a portion of the full picture. Combining the results helps because it aggregates different perspective on the same underlying phenomenon.

In summary, in both situations, combining three tests with a majority rule strictly improves diagnostic accuracy yet through different mechanisms. Although useful for building intuition, the distinction between these two sources of disagreement is hypothetical. Reality is likely to be a mixture of the two extreme situations, and it might be hard (if not impossible) to identify which factors are systematic versus stochastic. Fortunately, doing this is not needed to apply our approach, which takes the correlation structure observed in the data as its primitive. Our two examples are also simplistic in the sense that, in practice, accuracy is a trade-off between sensitivity and specificity (i.e., performance on both infected and uninfected populations). Furthermore, a systematic methodology is required to decide which tests to combine and how to interpret their results. The tools developed in this paper specifically achieve these objectives.

## Endnotes

[1] We note that standard Bayesian approaches might be used to interpret discordant outcomes in cases where priors are available and the testing accuracy (sensitivity and specificity) is known for each test. However, such approaches would not apply in settings where prevalence and accuracy must be estimated from noisy data and would not allow for efficiently optimizing which tests to combine.

[2] Among $\phi$ divergences, Van Parys et al. (2021) prove that uncertainty sets based on the Kullback–Leibler divergence are optimal in terms of out-of-sample disappointment. However, the resulting robust optimization problem here would be a mixed-integer conic optimization problem over the exponential cone. For tractability considerations, we have favored total absolute deviation, which leads to a simple mixed-integer linear program.

[3] Typically, more data are available for estimating specificity because condition-negative pre-COVID-19 freeze-dried samples are available (whereas sensitivity estimation requires condition-positive samples), so the confidence intervals on specificity estimates are smaller.

[4] The Wilson score method leads to CIs that are not centered around the empirical mean and uses a biased estimator instead, which has lower variance and overall lower mean squared error. This is a more appropriate method when the sample mean is close to the unit interval end points.

[5] For a single test, the AUC is equal to the arithmetic mean of the sensitivity and specificity.

[6] Figure EC4 in the e-companion decomposes these numbers for each heuristic rule separately.

[7] As an example, the specificity of an antibody test depends on its ability to (i) detect the biomarker of interest (i.e., an antibody that is developed in the human body in reaction to SARS-CoV-2) and (ii) rule out other "crossreactive" biomarkers (i.e., antibodies developed in the body in reaction to other viruses). In the case of SARS-CoV-2 antibody tests, prior coronavirus infections are the source of some of the potential crossreactive biomarkers. Antibody tests operate by using a piece of the virus to detect antibodies in a blood sample. Many tests focus on detecting antibodies to the spike protein of SARS-CoV-2. A recent *Lancet* article (Petherick 2020, p. 1102) notes that "[t]here is a lot hanging on the uniqueness of the spike protein. In terms of the specificity of serological tests in which it is used, the more unique it is, the lower the odds of cross-reactivity with other coronaviruses–false positives resulting from immunity to other coronaviruses … the potential for cross-reactivity really depends on whether the new tests select sections of the spike protein that are particularly distinct across coronaviruses."

[8] Referring back to the example provided in footnote 7, consider a population of individuals who each have crossreactive antibodies to three prior coronaviruses. A pattern similar to that shown in Figure A.1(b) can emerge on condition-negative samples if the SARS-CoV-2 protein segment targeted by each of the three tests overlaps with that in (a different) one of these three prior coronaviruses. For a similar motivation related to antigen testing, please see Corman et al. (2021, table 3), where the tests examined clearly differ in their ability to rule out crossreactive coronaviruses.

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